

Authentic matrix (or real)	For biological analysts, serum, urine, saliva, or stool are different classes of matrices. In food chemistry, when determining the total protein, fatty and starchy foods are classified as different, or drinking water and surface water is different for water controllers.
Surrogate matrix	Matrix used as a substitute for authentic matrix.
Neat solution	Water, reagents used for extraction or elution, etc.
Artificial matrix	Pooled and homogenized samples, material prepared by weighting when the composition of the authentic matrix is fully known, etc.
Stripped matrix	Specially prepared materials are free of impurities or endogenous chemicals. They are mainly used for biomedical analysis.

It can be assumed that the combined use of surrogate standard and/or surrogate matrix may induce bias. It is necessary to cautiously verify if their analytical behavior is comparable to authentic ones. At least four combinations of the above-defined quantification elements are possible, each having pros and cons as explained later. It is possible to categorize different quantification modes depending on the selected combination:

Quantitative	Calibrators are prepared with authentic analytes and an authentic matrix. The amount or concentration of the analyte may be determined and expressed as a numerical value in appropriate units. The final expression of the result can be absolute, as a single concentration value; non-absolute, as a range or above or below a threshold.
Semi-quantitative	Surrogate standards and matrix are used. Some authors consider semi-quantitative analyses the ones performed when reference standards or the blank matrix are not readily available.
Relative	Sample is analyzed before and after an alteration or compared to a control situation. The relative analyte concentration is expressed as a signal intensity fold change. It is ratioed to another sample used as a reference and expressed as a signal/concentration.

It must be clearly stated that it is impossible to strictly separate quantification from calibration since they are interdependent. According to the nature of the calibration standard used, which can be authentic or surrogate, and the matrix, which can be authentic, surrogate, neat, etc., different quantification strategies were

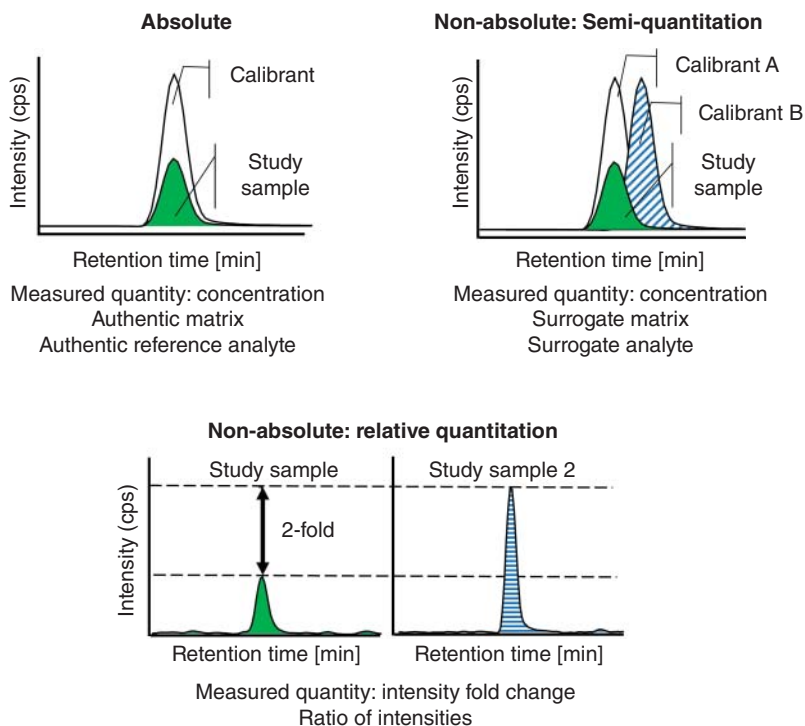


Figure 1.2 Schematic representation of absolute, semi, and relative quantification modes.

developed to obtain the effective calibration function. A schematic overview of the differences between principal quantification modes is summarized in Figure 1.2 and more extensively explained in the rest of the chapter.

1.1.3 Signal Pretreatment and Normalization

Nowadays, it is quite uncommon to use the analogic electrical signal output from the measuring instrument to build a calibration model. Digitalizing signals in modern instruments opened the way to many pretreatments, such as filtering, background correction, and smoothing. It is sometimes invisible to the analyst, although this can modify the method's performance. The outcome of many methods can be complex signals such as absorption bands or peaks in spectrophotometry or elution peaks in chromatography.

This raw information is not directly used as Y variable to build the calibration model; it is preprocessed. When dealing with absorption peaks, it is classic to select one or several wavelengths considered to be most informative. For instance, in biochemistry, protein concentration can be quickly estimated by measuring the UV absorbance at 280 nm; proteins show a strong peak here due to tryptophan and tyrosine residue absorbance. This can readily be converted into the protein concentration using Beer's law.

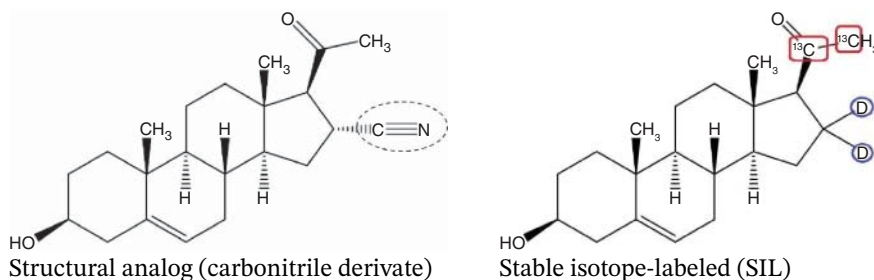
When obtaining poorly resolved absorption bands, as in near infrared spectroscopy (NIRS), the selection of one specific wavelength is difficult, and the use of a multivariate approach has been promoted. Many publications in chemometrics literature are addressing this issue. The multivariate calibration based on partial least-squares regression (PLS) has now become a routine procedure.

If the output signal is time-resolved, such as liquid or gas chromatographic peaks, they are always pretreated by an integrator. Initially, it was a separate device, but now it is included in the monitoring software. It can determine several parameters characterizing the elution peak, such as retention time at the highest point, skewness, peak height, but mainly peak area. The peak area is in the favor with analysts. But several publications demonstrated that for some methods, peak height is preferable to peak area and that when standardizing a method, the integration conditions must be carefully harmonized [2].

For some methods, such as MS-coupled methods, the measured response Y can strongly vary according to the detector performance, such as mass analyzer type, ionization modes, ion source parameters, system contamination, ionization enhancement or suppression due to the sample matrix effect, along with other operational variables related to the analytical workflow.

Thus, the analyte relative response is standardized to compare performance over time. A common operation is adding an internal standard (IS) to the study and calibration samples at fixed concentrations. For instance, two official inspection bodies advise evaluating the matrix effects when a complex surrogate matrix is used [3, 4]. For the latter, the Food and Drug Administration (FDA) suggests investigating the matrix effect by performing parallelism testing between linear calibration curves computed with the authentic and surrogate matrices. This method is not always effective, while parallelism statistical testing is conservative, i.e. depending on the data configuration significant difference may be considered nonsignificant and only applicable to linear models.

Conversely, the European Medicines Agency (EMA) provides full instructions on how to do it and recommends comparing the extraction recovery between the spiked authentic matrix and surrogate matrix used for the calibration, along with the inclusion of IS as an easy and effective method to correct biases between these two matrices. When the analyte and the IS are affected similarly during the analytical process, instrument signals can be correctly standardized. A comprehensive approach is proposed further using the method accuracy profile (MAP); it is also an effective approach to detect and control matrix effects.



Two main categories of IS, namely structural analogs and stable SIL, can be identified. The molecule of pregnenolone is used to exemplify this. The first category, visible on the molecule on the left, is related to compounds that generally share structural or physicochemical properties similar to the authentic analyte.

The second category, exemplified by the molecule on the right, includes stable isotopic forms of the analyte, usually by replacing hydrogen ^1H , carbon ^{12}C , or nitrogen ^{14}N with deuterium ^2H , ^{13}C , or ^{15}N , respectively. Obviously, using labeled IS requires the coupling to a mass spectrometer. Deuterated IS are widely used due to their lower cost. Still, their lipophilicity increases with the number of substituted ^2H , leading to differences in their chromatographic retention times with the corresponding authentic analyte. This phenomenon, known as *deuterium effect*, can also impact the instrumental response or behavior (e.g. the electrospray ionization process in MS) compared to unlabeled compounds.

Even if an increasing number of high-quality SIL are commercially available, they are limited to the most commonly used chemical compounds. When many analytes must be simultaneously quantified, the possibility of using one IS for multiple analytes should be carefully evaluated. For quantification purposes, using one IS per target compound is generally recommended when available because they are assumed to compensate for specific differences in matrix effect and extraction recovery between the calibration methodology and working samples.

To complete this rapid overview, when compatible with the analytical method, the use of standards linked to the International System of Units (SI) is a convenient means of standardizing the instrumental response and correcting the overall variation in the measurement process resulting from diverse sources of uncertainty, such as sample preparation or interfering compounds, also known as the matrix effects. The absolute instrumental response is then normalized as a response ratio:

Normalized response ratio

$$Y = \frac{Y_A}{Y_{IS}} \quad (1.1)$$

In this formula, Y_A and Y_{IS} are the responses obtained with the analyte and the IS, respectively. This formula gives a relative instrumental response but does not consider the respective concentrations. To be more in harmony with Figure 1.1, Y_{IS} is equivalent to Y_c . This new notation is used because the IS is a particular example of a compound used for calibration.

The influence of signal preprocessing, such as peak integration, was experimentally demonstrated during an interlaboratory study on determining fructose, maltose, glucose, lactose, and sucrose in several foods by liquid chromatography [5]. A specific experimental design was developed to achieve this demonstration. Participants were requested to send their results calibrated as both peak heights and areas. Considering the mean values obtained with the two approaches, differences ranged from -18% up to $+5\%$. This indicates that trueness may be affected by the quantification mode. Precision, expressed as the reproducibility variance, was computed using both sets of results.

More details about this common parameter of precision are given in Section 3.2.1. In Figure 1.3, a subset of interlaboratory results is reported. Food types are indicated

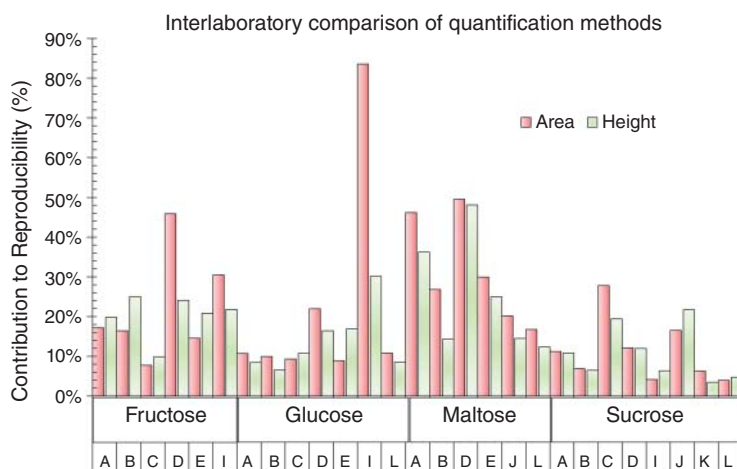


Figure 1.3 Contribution to the reproducibility of two quantification methods in liquid chromatography of saccharides.

by an uppercase letter ranging from A to L; they are saccharide-containing processed foods, such as soft drinks, baked foods, or candies. Precision for peak area appears as vertical red bars and peak height as light green bars. The role of the signal processing method is expressed as a relative contribution to the reproducibility variance. The contributions and their differences are sometimes ridiculously small, such as fructose in food C where it is below 10%. But sometimes very impressive, such as glucose in food I. If some food is not present on the diagram, the analyte was not detected. For instance, *L* is a chocolate bar that contains no fructose. Peak area is not always the best way to quantify the analyte. In the publication, an explanation is given why the discrepancies exist. It mainly depends on the resolution of peaks and their relative values.

Detecting a peak beginning and end is a contingent subject and a source of uncertainty for the surface integration, as explained in Section 4.1.2. Finally, integrator settings can be used to optimize the integration algorithm and accordingly influence the global performance of the method.

1.2 Calibration Modes

Two major calibration modes are used in laboratories, namely:

External calibration (EC)

A calibration curve is established independently from the working samples, whatever the calibrant nature and preparation. A single calibration function is used to quantify many samples. This is the most classical procedure, and several variants exist.